Decision Log — Knotted Coil Designer

Dated entries: what was decided, why, who decided it, and what the alternatives were. Newest first.

2026-10-20 — Design v1.5: quick B1 view

- **Request (Reza):** “I just want a quick visualization that’s reasonable. We will do simulations later.”
- **What was built:**
  - quasi-static Biot–Savart field of the solved wires;
  - a placeholder head surface (helmet − 10 mm) and axial, coronal or sagittal slices;
  - elements combined as $\sqrt{\Sigma |B1_i|^2}$, or a single element;
  - colour in dB relative to the maximum shown.
- **Check:** V8 pre-registered (git ff1a40e). Its closed form is DERIVED, not cited.
- **Speed:** about 0.3 s for the surface view (16 elements).
- **E-fields:** still excluded. A partial (inductive-only) E map would mislead. Recommendation: export the wire paths to a full-wave simulator.

2026-10-20 — Validation run 5 (v1.4): V7 FAIL at 1 of 360 points

- **Failing point:** T(2,5), m = 16, β = 1.5. The drawing showed 6 crossings instead of 5 at 25,600 points.
- **Cause:** a duplicate hit at the 54° crossing, where the strands meet at a grazing angle. The count is exactly 5, in the right positions, at 51,200, 102,400 and 204,800 points.
- **Fix:** point-count ceiling raised from 25,600 to 102,400. Criterion unchanged. Re-run as run 6.

2026-10-20 — Design v1.3: petal sharpness

- **Request (Reza):** “Add a petal-sharpness control that narrows each lobe, with the same topology checks.”
- **Implementation:** outward lobe profile raised to a power m, r = R(1 + β((1 + cos qϕ)/2)^m). Tips and base are unchanged; the lobes narrow.
- **Range:** 1–16. Not applied to the symmetric v1.0 shape.
- **Lobe aspect ratio:** length ÷ width at half length is now reported.
  - Example, cinquefoil at β = 1.5: 0.68 (m = 1), 1.31 (m = 4), 2.05 (m = 10).
- **Topology check V7:** pre-registered in Design §7 (git 30d6fa1) before it was run.
- **V7 RESULT (validation run 4): FAIL.**
  - **Passed across the whole grid:** the trefoil T(2,3).
  - **Failed at m ≥ 12:** T(2,5) and T(2,7).
  - **Failed from m ≈ 3–6:** the 3- and 5-winding knots.
  - **Mechanism:** extra crossings in the flat drawing, in pairs (e.g. 10 instead of 8). The power law flattens the base between petals, so neighbouring strands run almost on top of each other and cross back and forth.

    - **Knot type:** Claude argues it is probably unchanged, because the 3D curve still lies on an embedded torus. **Not verified computationally.** The diagram is still not clean: it has extra wire crossings and long stacked runs.
    - **Response:** v1.4 replaces the power law with an angle warp.
- **v1.4 (git 7d7b6bf):** sharpness warps the lobe angle, $\psi = q\phi + (1 - 1/m)\cdot\sin(q\phi)$, and the same ψ drives both radius and lift.
    - Claude's argument: crossings stay exactly $q(p-1)$, with the same pattern.
    - Probes showed residual mismatches only at $m = 16$, $\beta = 0.1$. They were confirmed as discretisation artefacts: they vanish at 2,000 points for T(3,2) and 16,000 for T(5,6).
    - **Numerical changes:**
        - a grid-accelerated crossing and clearance search (identical to brute force on 108 crossing cases and 6 clearance cases);
        - a point-count rule requiring stability over two doublings (max 25,600 points).
    - **Trade-offs seen in probes (not validated):**
        - Aspect ratio gains are larger for 2-winding knots (T(2,5) at $\beta = 1.5$: 0.68 → 1.25 from $m = 1$ to 16) than for 3–5-winding knots (T(3,4): 0.36 → 0.66).
        - The crossing lift needed grows with m, up to about 9.8 mm for T(5,6) at $m = 16$. That is likely impractical.
- **Bend radius:** tightens with m, e.g. 3.9 mm at $m = 10$ for that cinquefoil. The app flags this against the wire's minimum bend radius.

2026-10-20 — Design v1.2: bloom extends lobes (outward shape)

- **Request (Reza):** bloom should lengthen the lobes, with aspect ratio rising with bloom, rather than pinch part of the loop.
- **Cause of the pinch:** the v1.0 formula $R(1 + \beta \cos q\phi)$ moves the lobe tips out *and* the waist between lobes in by the same amount. At high β the waist collapses toward the centre.
- **Change:**
    - New default: $R(1 + \beta(1 + \cos q\phi)/2)$. The base circle stays fixed and the lobes extend outward only (lobe length = βR).
    - β_max raised to 3.0.
    - The v1.0 "symmetric" shape is kept as an option.
- **Alternative considered:** an extra "petal sharpness" exponent to narrow each lobe as well. Not built; the length change alone makes aspect ratio rise with β.
- **Side effect:** the bloom floor doubles to $c / (R \sin(\pi/p))$. At R = 30 mm with the placeholder wire, that is 0.05 for $p = 2$ and 0.058 for $p = 3$.
- **Pre-v1.2 results:** validation run 2 and the worst-pair comparison used the symmetric shape. Their conclusions do not automatically carry over.

2026-10-20 — DECIDED: "decoupled" = nearest ring (Design v1.1)

- **Decided by:** Reza, choosing between the frozen footprint-overlap rule and the nearest-ring rule.

- **What changed:**
  - Solver default is now `target = 'ring1'`.
  - The app lists the nearest-ring verdict first.
  - The footprint verdict (v1.0 rule) is still computed and shown for every result.
  - Design.md changelog updated.
- **Alternatives:** keep the footprint rule, with its ill-posed objective; or a Voronoi-adjacency definition (not built).

2026-10-20 — Worst-pair term implemented; made default by pre-registered rule

- **Requested by:** Reza: "Implement the worst-pair term as a smooth stand-in (soft-max or sum of k^4), compare against the current solver on the same seeds, and report both."
- **Implementation:** per-pair cost $w\cdot(k^2 + \lambda_{max}\cdot k^4/thr^2)$, with $\lambda_{max} = 1$ (JUDG, set before testing, not tuned).
- **Protocol:** `validation/COMPARE_worstpair_PROTOCOL.md`, committed before running (git 6d96721). 35 paired runs:
  - T(2,3), T(2,5), T(3,4) at N = 16 and 32, seeds 1–5, fast solver;
  - plus T(2,3) at N = 16, seeds 1–5, full solver.
- **Result:** all three criteria met.
  - B (with the term) had the lower worst nearest-ring |k| in 26 of 35 runs.
  - Median paired ratio B/A = 0.84.
  - Coverage violations: 0.
- **Applied mechanically:** default is now `lambdaMax = 1`.
- **Trade-off shown by the secondary outcomes (reported, not part of the rule):**
  - median nearest-ring |k| rose from 0.0052 to 0.0119;
  - median count of pairs ≥ threshold rose from 17 to 22.
  - In short, the term lowers the single worst pair by spreading moderate coupling over more pairs.
  - Second-ring worst |k| was essentially unchanged (0.066 vs 0.067).
- **Also seen in both arms:**
  - No N = 32 run met the threshold for all nearest-ring pairs.
  - Results vary strongly with the random seed (same configuration, A's worst |k| ranged 0.010–0.042).
  - The solver is a weak link. Restarting from several seeds and keeping the best would be a cheap improvement (not built).
- **Open for Reza:** which matters more for his array, the worst pair or the typical pair. If the typical pair matters more, set the weight to 0; the app exposes it.

2026-10-20 — Deviation found: λ_max term not implemented (resolved by the entry above)

- **What happened:** Design §5.1's cost includes $\lambda_{max}\cdot\max|k|$. The solver as built (git b21c784) omits it. Claude found this while editing §5.1 for v1.1 and disclosed it the same day.
- **Effect:** the solver can trade one bad pair for many good ones. The worst pair is always reported but not specifically targeted.

- **Options:**
  - implement it, e.g. as a smooth stand-in such as Σk^4 or a soft-max, because a hard max is awkward for an element-by-element local search;
  - or amend the Design to drop it.
- **Status:** awaiting Reza.

2026-10-20 — Field maps (B1 / E) requested; not built yet

- **Request (Reza):** a UI toggle to visualise E-fields.
- **Claude's assessment:** a quasi-static wire model can compute the **magnetic field** properly, using Biot–Savart, and that can be checked against the analytic on-axis field of a loop. It can only give the **inductive part of E** ($-\partial A/\partial t$). The charge-driven part from capacitors and gaps, and the effect of the head, are missing, so an E map would be incomplete in a misleading way.
- **Status:** outside frozen Design v1.0 scope (§1, §8). **Awaiting Reza's decision.**

2026-10-20 — In-plane rotation as a solver option

- **Decided by:** Reza, who asked for rotation as a decoupling lever.
- **What changed:** rotation was already a free variable per element. It is now exposed as `rotation` = free / locked (one shared angle) / fixed.
- **Exploratory observation, one seed, N = 16, trefoil β = 0.4, nearest-ring target:** worst nearest-ring |k| was 0.010 with free rotation, 0.040 locked and 0.045 fixed.

2026-10-20 — Neighbour definition: frozen rule kept as default, 'ring1' added (superseded by the v1.1 entry above)

- **Finding:** a trefoil at β = 0.4 nulls only at a centre spacing of about 1.14 Rf, which is heavy overlap. At that spacing, second-ring elements (about 1.97 Rf in a hexagonal layout) still overlap.
- **Consequence:** under the frozen Design §5.1 definition (footprints overlap), the solver must null pairs that cannot be nulled together, and the problem is badly posed.
- **Option added:** `target = 'ring1'` decouples only the nearest ring (closer than 1.35 × the pair-null spacing).
- **Results (16 elements, exploratory):**
  - nearest ring: worst |k| ≈ 0.008–0.010;
  - second ring: |k| up to about 0.08.
- **Default is unchanged** (frozen definition). Every result reports **both** verdicts.
- **Decision needed from Reza:** which definition should be the default, i.e. which one "decoupled" should mean.

2026-10-20 — Validation runs 1 and 2: V6 FAIL (recorded), causes and fixes

- **Run 1** (git 9f89d91): V6 failed for three separate reasons.

1. **Physics:** the bloom floor is above β = 0.02 for the placeholder wire (1 mm + 2 × 0.25 mm insulation) at R = 30 mm. Near the knot, the strands sit on a thin tube of in-plane radius βR. The p strands can be at most $2\beta R \cdot \sin(\pi/p)$ apart, so with c = d_w + 2t_i the minimum buildable bloom is $\beta \geq c / (2R \sin(\pi/p))$. That gives

0.025 for p = 2, 0.029 for p = 3 and 0.043 for p = 5. This is Claude's derivation; the numerical lift solver agreed (feasible at 0.03, 0.03 and 0.045).

2. **Discretisation:** 300 points missed one crossing of T(2,5) at β = 0.9 (600 or more resolve it). That shape also has a 0.08 mm bend radius, so it cannot be built.
3. **Harness bug:** the reference over/under pattern was taken at β = 0.02, which was itself the degenerate case.

- **Fixes:**
  - adaptive point count (`knot.adaptiveN`);
  - reference pattern taken at β = 0.4;
  - bloom floor shown in the UI.
  - The **criterion is unchanged.**
- **Run 2:** V6 still FAILS, now *only* at β = 0.02, for every knot. That is the physical floor.
- **Post-hoc check PH2 (informational, not a verdict):** topology is preserved from the floor upward for all catalogue knots.
- **Decision needed from Reza:** whether to change the β_min default to the wire-dependent physical floor. That is a Design change.

2026-10-20 — V3 reference is UNVERIFIED

- **Status:** Roemer et al. 1990 is paywalled; a subagent tried the Wiley PDF, Semantic Scholar, OpenAlex, Europe PMC and the Internet Archive.
- **Closest primary text read:** Roemer & Edelstein, US 4,825,162 (1989), "D approximately equal to 0.75d". This is PARAPHRASE-ONLY, read from Google Patents OCR text.
- **V3 verdict:** NOT EVALUABLE. Our computed value is 0.7604 × diameter (exploratory comparison: +1.4% from the patent's approximate value).
- **To resolve:** Reza retrieves the paper through McGill library access.

2026-10-20 — Kernel softening between elements

- **Choice:** wires of *different* elements use a $1/\sqrt{(r^2 + \varepsilon^2)}$ kernel with ε = d_w + 2t_i, so crossings between elements are treated as separated by the real centreline gap. This avoids having to model which wire goes over at each crossing.
- **Alternative:** explicit over/under layers per crossing.
- **Status:** sensitivity to ε not yet tested.

2026-10-20 — Design.md v1.0 frozen

- **Decided by:** Reza approved all §11 defaults.
- **Commit:** git 39a4e83; this commit pre-registers V1–V6.